According to the problem, the difference between two timings

$$= 4-(-5)$$
$$= 4+5 = 9 \text{ minutes}$$
$$= \frac{9}{60} \text{ hour (changing minutes to hour)}$$

Given, $T_1 - T_2 = \frac{9}{60}$

$$\frac{x}{35} - \frac{x}{50} = \frac{9}{60}$$
$$\frac{10x - 7x}{350} = \frac{9}{60}$$
$$\frac{3x}{350} = \frac{9}{60}$$
$$x = \frac{9}{60} \times \frac{350}{3}$$

The distance to his office $x = 17\frac{1}{2}$ km.

Exercise 3.7

1. Fill in the blanks:

(i) The solution of the equation $ax+b=0$ is_____.

(ii) If a and b are positive integers then the solution of the equation $ax=b$ has to be always_____.

(iii) One-sixth of a number when subtracted from the number itself gives 25. The number is_____.

(iv) If the angles of a triangle are in the ratio 2:3:4 then the difference between the greatest and the smallest angle is _____.

(v) In an equation $a + b = 23$. The value of a is 14 then the value of b is_____.

2. Say True or False

(i) "Sum of a number and two times that number is 48" can be written as $y+2y = 48$

(ii) $5(3x+2) = 3(5x-7)$ is a linear equation in one variable.

(iii) $x = 25$ is the solution of one third of a number is 10 less than the original number.

3. One number is seven times another. If their difference is 18, find the numbers.

4. The sum of three consecutive odd numbers is 75. Which is the largest among them?

5. The length of a rectangle is $\frac{1}{3}$ of its breadth. If its perimeter is 64m, then find the length and breadth of the rectangle.

6. A total of 90 currency notes, consisting only of ₹5 and ₹10 denominations, amount to ₹500. Find the number of notes in each denomination.

7. At present, Thenmozhi's age is 5 years more than that of Murali's age. Five years ago, the ratio of Thenmozhi's age to Murali's age was 3:2. Find their present ages.

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